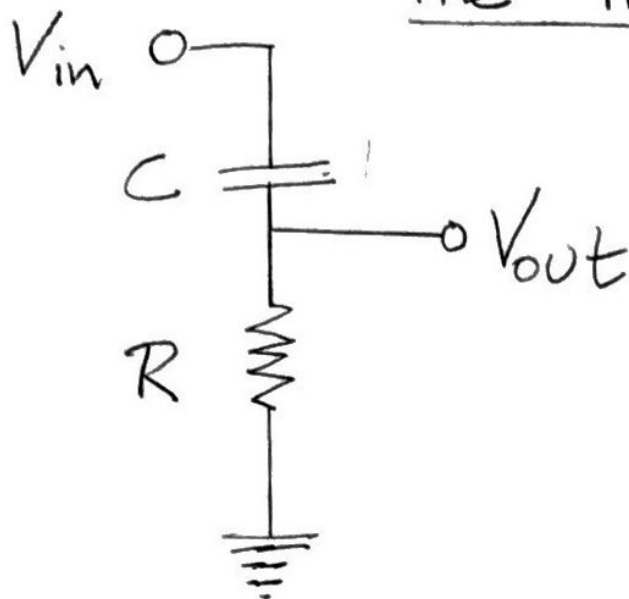


The High-Pass Filter



As in problems 7 and 8 on the last problem set, weirdly, for input voltages of a definite frequency, the capacitor can be thought of as having a resistance at 50Hz

$$R' = \frac{1}{2\pi f C}$$

$$= \frac{1}{2\pi \cdot 50\text{Hz} \cdot 100\text{nF}} = 3183\Omega$$

$$\approx 30\text{k}\Omega \text{ (rounded to one sig fig)}$$

And at 5000Hz

$$R'' = \frac{1}{2\pi f C} = \frac{1}{2\pi \cdot 5000\text{Hz} \cdot 100\text{nF}} \approx 300\Omega$$

This time the formula that relates V_{in} and V_{out} is

$$(a) V_{out} = V_{in} \frac{R}{\sqrt{R^2 + R'^2}} \quad \begin{array}{l} \text{use } 10V \\ \text{50Hz} \\ \text{case} \end{array}$$

=

Use

$R = 3000 \Omega$
in (a) and (b)

$$(b) V_{out} = V_{in} \frac{R}{\sqrt{R^2 + R''^2}} \quad \begin{array}{l} \text{again, use} \\ 10V \end{array}$$

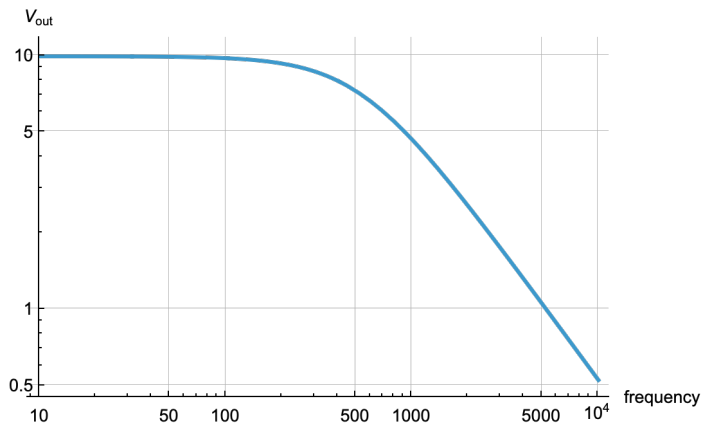
=

If all went well, you'll get that the high-frequency input was not attenuated as much as the low-frequency input. "High Pass."

```
In[37]:= r = 3000;
vIn = 10;
c = 100 × 10-9;
```

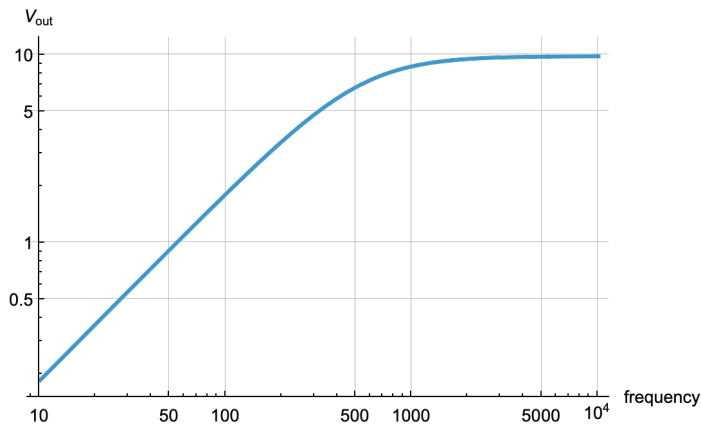
Frequency Response in Low-Pass Arrangement

```
In[40]:= vOutLowPass[f_] := vIn  $\frac{\frac{1}{2 \text{ Pi } f c}}{\text{Sqrt}[r^2 + \left(\frac{1}{2 \text{ Pi } f c}\right)^2]}$ 
In[41]:= LogLogPlot[vOutLowPass[f], {f, 10, 10 000},
GridLines → Automatic, AxesLabel → {"frequency", "Vout"}]
Out[41]=
```



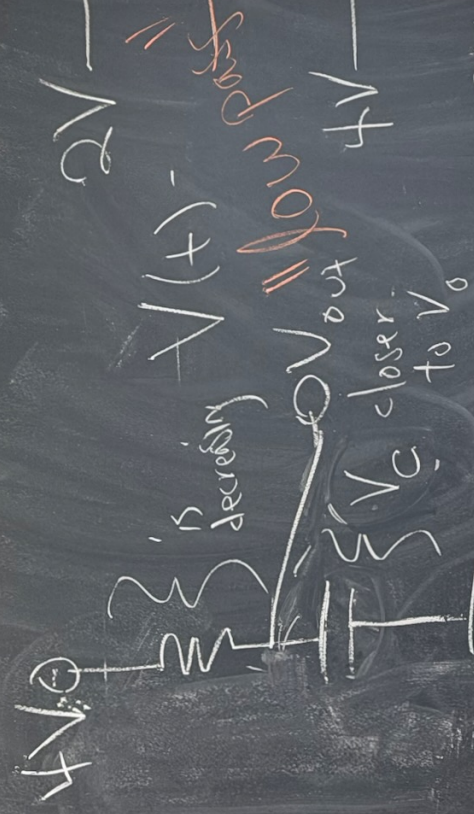
Frequency Response in High-Pass Arrangement

```
In[42]:= vOutHighPass[f_] := vIn  $\frac{r}{\text{Sqrt}[r^2 + \left(\frac{1}{2 \text{ Pi } f c}\right)^2]}$ 
In[43]:= LogLogPlot[vOutHighPass[f], {f, 10, 10 000},
GridLines → Automatic, AxesLabel → {"frequency", "Vout"}]
Out[43]=
```



$$V(t) = V_0 (1 - e^{-t/\tau})$$

$$\tau = -\tau \ln \left(1 - \frac{V(t)}{V_0} \right)$$



$$\tau = RC$$

